Another byproduct of this fact is that some of the old rules of investing—namely that for the same level of investment volatility or risk (as measured by standard deviation), one is better off with a higher return—may no longer apply. One of the fundamental axioms of modern portfolio theory and current investment management is that investments can be ranked solely on the basis of their mean (average) return and variance (volatility).

Stated differently, there is a strong relationship between investments that exhibit a high degree of return variability (that is, they fluctuate a lot) and their long-term growth rate. Variability tends to be measured by standard deviation. This is illustrated in Figure 6.5, where the upper portion of the curve, which is often called the investment frontier, moves higher for greater values of standard deviation. In general, investors are counseled to never accept lower return (for example, 8% compared to 7%) from a product, if the statistical fluctuations of the two products are identical. In retirement, this may no longer be the case since the traditional measures of risk, the daily fluctuations, may not capture the sequence of return risk that I described.

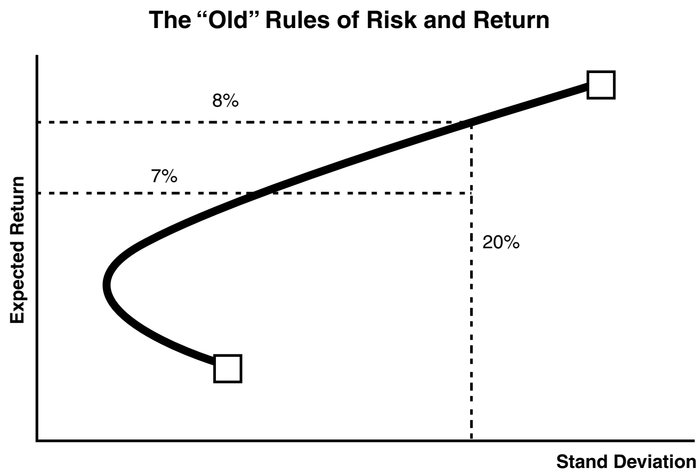

Figure 6.5 Why would you give up 100 basis points?

Source: Based on work by H.M. Markowitz, "Portfolio Selection," *Journal of Finance*, 1952, 7(1):77-91.

Anyway, the technical details of these regressions can get somewhat numbing but the bottom line here is that the sequence-of-investment-returns effect is real. You must protect yourself against